

Preserved Discrimination but Impaired Interpolation in hV4 Identifies the Correctable
Distortion in Colour Vision Deficiency

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Abstract

Placeholder abstract. Replace this with your actual abstract text. This study investigated neural color representations in individuals with color vision deficiency (CVD) using functional MRI and computational modeling approaches.

Keywords: color vision deficiency, fMRI, shared response model, forward encoding model, representational similarity analysis

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Introduction

Color vision deficiency (CVD) affects approximately 8% of males and 0.5% of females of Northern European descent (Bosten, 2019) and arises from altered spectral sensitivity of the L- or M-cone photopigment, driven by X-linked opsin-gene polymorphisms (Neitz & Neitz, 2011; Deeb, 2005). In anomalous trichromats the deviant cone’s peak sensitivity is displaced by approximately 2–12 nm from its normal value (Stockman & Sharpe, 2000). This shift reduces the L–M cone-opponent signal rather than abolishing it. The peripheral consequences — reduced red–green discrimination, category confusions, and elevated just-noticeable differences along the L–M axis — are the classical phenotype captured by the Ishihara test (Ishihara, 1917) and by subsequent psychophysics (Bosten, 2019).

CVD is not an on/off loss: downstream cortical color machinery remains intact and receives a quantitatively reweighted but qualitatively similar input. This makes CVD a paradigmatic example of a condition in which a high-dimensional neural representation is not absent but *structurally distorted*. The scientific and translational question is therefore not whether color information survives the retina — it manifestly does — but whether, and how, its cortical geometry is reshaped at the level of the individual. This question remains open, and an individualized assistive filter must answer it before it can be designed.

CVD correction filters in current use fall into two structurally related families. Hardware filters — dichroic notch lenses such as the EnChroma line — attenuate spectral bands near the L–M crossover to widen the post-receptoral L–M signal. Software filters operate at the display: Brettel–Viénot–Mollon simulation (Brettel, Viénot & Mollon, 1997) and the parametrised Machado retinal model (Machado, Oliveira & Fernandes, 2009) forward-project a stimulus onto an anomalous retina. Their Daltonization inverses, including recent learned reformulations (Akalin et al., 2025), redistribute the lost contrast into channels the user can still see. Despite this

diversity, the family shares two structural commitments: optimisation against a *population-average* retinal model, with the same transformation applied to every user.

Attempts to escape this average-user ceiling have predominantly taken one route: tuning a retinal-model parameter — the simulated L- or M-cone shift in Machado-class models — to a user’s responses on plate-style or appearance-matching tests. The route is constrained at its target: its optimisation criterion is the user’s appearance *report*, not the cortical representation that downstream tasks read out.

This distinction matters because long-term cortical adaptation in anomalous trichromats reshapes hue appearance independently of the retina (Tregillus et al., 2021). Appearance-matched calibration therefore inherits whatever compensation the cortex has already imposed. A device tuned to undo only the retinal component then re-renders the input *as if* that cortical compensation did not exist. Consistent with this account, clinical validation of consumer color-enhancing CVD-correction glasses reports at most modest changes in chromatic discrimination thresholds in congenital red–green deficiencies (Patterson et al., 2022), and recent behavioural work likewise shows that population-average filters change the appearance of color without substantively shifting discrimination thresholds at threshold contrast (Somers, Franklin & Bosten, 2024), while the underlying CVD phenotype varies markedly even within a single diagnostic category (Bosten, 2019). What is missing from this landscape is a filter derived from, and validated on, the *user’s own* neural color geometry.

Cortical color representation is a distributed, hierarchical computation (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals. Higher-order regions, in particular hV4 (Brouwer & Heeger, 2009; Bannert & Bartels, 2018), support a population code for continuous hue that can be read out by forward encoding models (Brouwer & Heeger, 2009; Parkes, Marsman, Oxley, Goulermas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). Cross-subject shared-response modelling has recently been validated for healthy color decoding (Bannert & Bartels, 2025), providing a methodological scaffold that the present clinical extension adopts.

Behavioural work converges on the conclusion that anomalous trichromats do not merely discriminate less well; they compensate. Long-term cortical adaptation reshapes hue appearance (Tregillus et al., 2021; Boehm, MacLeod & Bosten, 2014; Webster, 2015), and hue-scaling experiments in anomalous trichromats ($n = 10$ AT vs. 26 NT) reveal a consistent 21.4° rotation of the blue–yellow response peak — a phase shift of the half-rectified cosine fit to scaling responses across the 360° stimulus circle (NT peak 127.5° vs. AT peak 106.1° ; $t(34) = 5.95$, $p < 0.001$) (Emery, Volbrecht, Peterzell & Webster, 2021; Isherwood, Joyce, Parthasarathy & Webster, 2020). These results locate the key variance not at the retina alone but at the retinal–cortical interface.

Yet three concrete gaps remain. (*Gap 1*) Existing neural studies of color characterise population-mean geometry in healthy observers (Brouwer & Heeger, 2009; Bannert & Bartels, 2018; Kuriki et al., 2015) or, in CVD, group means at the level of activation magnitude. To our knowledge — based on PubMed and Google Scholar searches combining “CVD” / “colour vision deficiency” / “anomalous trichromacy” with “fMRI”, “forward encoding”, and “individual differences” — none has inverted the high-dimensional cortical signal into an interpretable, individualized distortion field. (*Gap 2*) CVD neural work has relied principally on classification accuracy, which is a sufficient statistic for category identity but discards the continuous relational structure (Brouwer & Heeger, 2013) that downstream perceptual and translational tasks actually demand. (*Gap 3*) Behavioural evidence for cortical compensation has not been translated into a concrete criterion for choosing between retinal-level and cortical-level correction strategies — an omission that directly limits filter design.

A filter for a continuous-hue condition must operate on continuous-hue representation, not on categorical labels. This observation motivates a complementary pair of cross-validation schemes on the same forward encoding model. Leave-one-run-out (LORO) cross-validation probes whether the 8 trained hues can be *classified* from held-out runs. It measures category-level discriminability and is the minimum requirement for a filter to matter: if the target cortex fails to discriminate the displayed colors, no stimulus-space adjustment has leverage on which to act.

Leave-one-colour-out (LOCO) cross-validation, by contrast, probes whether the encoder *interpolates* from its 7 training hues to a held-out hue. It measures the integrity of the continuous manifold on which any hue correction must act.

A joint pattern is therefore predicted and empirically required: LORO *preserved* in CVD (the filter precondition) together with LOCO *impaired* (the filter target). Finding only the first would make a filter redundant; finding only the second would preclude any stimulus-space remedy. Finding both together defines the regime in which individualized corrective filters are simultaneously warranted and tractable, and in which the per-hue LOCO vulnerability vector — the subject-specific signature of where the continuous geometry fails — becomes the phenotype to be inverted.

We test whether *individual hV4 colour geometry* carries person-specific distortion through a single-case fMRI characterisation of two CVD adults (one moderate deutan, one moderate-severe protan) under Crawford–Howell single-case inference (see Methods), alongside a normative reference of seven healthy controls. Findings specific to sub-09 — the single protan participant in our cohort — are reported as exploratory single-case observations requiring independent replication.

1. Does the continuous cortical hue geometry in CVD carry

individually-structured distortion? We use the LORO/LOCO dissociation on hV4 forward encoding to distinguish preserved discrimination from impaired interpolation, and extract a per-hue vulnerability vector $\mathbf{v} \in \mathbb{R}^8$ as each individual’s geometric fingerprint.

2. Can that distortion be expressed in a small number of physiologically interpretable parameters, and does an independent geometric criterion corroborate the fit? We infer 1–2 DOF parameters from $\mathbf{v}$ under three mechanistic model families (retinal Machado, retinal–cortical R+C, cortical 2-component) with closed testing. We then cross-check the winning fit against the pairwise SRM-aligned Δ RDM and against the cross-cohort S-cone compensation angle described above.

3. Does the pre-image of the fitted distortion constitute a testable corrective filter, and do retinal and cortical hypotheses agree on what to display? We compute the per-hue pre-image of each fit and assess bijectivity. Retinal and cortical fits converge on *who* is distorted but diverge on *how* to correct — generating a falsifiable behavioural prediction to be tested in a same-day 2AFC discrimination experiment.

We hypothesized that (i) categorical hue discrimination at hV4 is preserved in both CVD individuals while continuous-hue interpolation is selectively impaired along S-cone-intermediate hues; (ii) the resulting per-hue vulnerability profile is captured by a small number of physiologically interpretable parameters anchored on the S-cone opponent axis; and (iii) the pre-image of these parameters yields a stimulus-space correction filter whose direction differs between retinal-level and cortical-level model fits. Together, the three steps yield a per-individual filter whose input (cortical signal), intermediate parameters, and output (stimulus shift) are each testable in a single CVD user. We note that per-subject statistical significance under label permutation cannot, by itself, establish CVD-specificity: a 2-component fit to the 8-hue vulnerability profile reaches nominal significance for all 7 HC controls under the same procedure (Supplementary). Case-level specificity therefore rests on joint anchors — HC parameter-norm bracket, directional consistency with cone-physiology, and (pending) Phase-3 behavioural validation — rather than on per-case p -values.

Methods

The analysis follows three stages. First, the fMRI neural signature of each CVD participant’s hue-processing deficit was characterised in two ways: (1) leave-one-color-out (LOCO) decoding at hV4 yielding a hue vulnerability profile $\mathbf{v} \in [0, 1]^8$ (see the Interpolation decoding (LOCO) section), and (2) representational geometry analysis quantifying pairwise hue dissimilarity structure via Δ RDM (see the Δ RDM analysis section); both serve as fitting criteria for the distortion model. Second, a two-component cortical distortion model fits these neural observations by simulating

the joint effect of a retinal cone shift and a cortical orientation gain on HC-trained neural representations (see the Two-component distortion model section). Third, the fitted distortion is analytically inverted to yield a stimulus-space pre-image filter that maps the distorted hue coordinates back towards the HC percept (see the Stimulus-space filter section).

Participants

Thirteen volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying ≥ 9 plates correctly were classified as healthy controls (HC; $n = 7$, 3 female, age 22.7 ± 2.5 years). Two male participants scoring below this threshold were classified as CVD: sub-08 (deuteranomalous, 5/14 plates corrected) and sub-09 (protanomalous, 7/14 plates corrected). Four additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified t -test; no population-level inference is made.

Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE $L^*a^*b^*$ space at equal angular spacing (0° – 315° , 45° steps; $L^* = 75$, chroma = 40) (Commission Internationale de L'Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ($a^* = b^* = 0$) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design to sustain attention and fixation. A circular patch (1.5 s per stimulus) was shown in pseudo-random order optimised with Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants pressed a button upon detecting two consecutive 'K's within a rapid serial letter stream presented above the grating patch. Each run contained eight repetitions of all eight chromatic stimuli and the gray filler; inter-stimulus intervals were

drawn from $\{3, 4.5, 6\}$ s. Each session comprised six runs (≈ 7 min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

CVD participants additionally completed a second scanning session for filter evaluation (see the Behavioral concordance section).

MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009) (TR 1.5 s, TE 30 ms, FA 75° , $2 \times 2 \times 2$ mm, 96×80 matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996). T1w-to-MNI normalisation used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at $> 50\%$ probability, intersected with each participant’s BOLD brain mask (voxel counts: V1 655 ± 214 ; V2 451 ± 145 ; V3 103 ± 29 ; hV4 63 ± 22 ; mean $\pm$ SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1 328 ± 107 ; V2 226 ± 73 ; V3 52 ± 15 ; hV4 32 ± 11 voxels). Second, the estimated HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a $6 \times 8 \times V$ array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimising $\|X_{\text{ref}} - XQ\|_F$ (scaling prohibited).

Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \quad (1)$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. Reduced dimensions were chosen by mean-rank aggregation across seven-fold leave-one-subject-out (LOSO) cross-validation, using three SRM-quality metrics (RDM split-half reliability, cross-subject RDM correlation, and reconstruction error): V1 $k = 4$; V2 $k = 4$; V3 $k = 3$; hV4 $k = 3$.

Forward encoding model

Both discrimination (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a $F = 6$ -dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$, where $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k (preferred hues spaced equally at 60° ; Brouwer & Heeger, 2009). A weight matrix $W \in \mathbb{R}^{F \times V}$ is estimated by ridge regression with cross-validated regularization (RidgeCV; Pedregosa et al., 2011) to predict voxel amplitudes from channel activations: $\hat{X} = CW^\top$. Decoding inverts the encoding direction: for a held-out voxel pattern $\mathbf{x}$, channel activations are reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$, and the candidate hue whose basis

vector has the highest Pearson correlation with $\hat{\mathbf{c}}$ is returned as the decoded colour. Alternative decoders evaluated during model selection are described in Appendix .

Discrimination decoding (LORO)

Categorical discrimination was assessed with leave-one-run-out (LORO) cross-validation in SRM-aligned space using the forward encoding model (see the Forward encoding model section). For each of the six runs, W was estimated on the remaining five runs using mean-subtracted SRM projections, and each held-out run's 8 colour patterns were decoded by nearest-neighbour matching. Primary discrimination performance was the 8-class exact accuracy (acc_exact; chance = $1/8 = 0.125$), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject generalization scores: the encoding model trained on each HC participant's SRM-aligned data was applied to every other participant's aligned patterns, yielding 21 unique HC-to-HC and 14 HC-to-CVD pairwise generalization scores. Within-ROI single-case tests used the Crawford and Howell (1998) modified t -test (one-tailed, lower).

Interpolation decoding (LOCO)

Continuous hue interpolation was assessed with leave-one-color-out (LOCO) cross-validation using the same forward encoding model (see the Forward encoding model section). Each of the 8 hues was held out in turn; W was estimated on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

Adjacent accuracy. The primary LOCO metric was adjacent accuracy (adj_acc): the proportion of predictions that fell within ± 1 hue step (45°) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a uniform random predictor = $3/8 = 0.375$; exact-accuracy chance = $1/8 = 0.125$). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4, p -values were obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified t -test (one-tailed, lower), respectively.

Vulnerability profile. The 8-dimensional vector of per-hue `adj_acc` values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. For group-level model fit, we report Spearman ρ between the distortion-model-predicted and observed $\mathbf{v}$ (see the Two-component distortion model section). Per-hue group comparisons (Figure 2C) used the Crawford and Howell (1998) modified *t*-test (one-tailed, uncorrected). Effect sizes are reported as Hedges’ *d* (Hedges & Olkin, 1985) with the Crawford–Howell correction for small-*n* comparisons.

Δ RDM analysis

To characterise the geometric basis of LOCO impairment we computed two measures in SRM-aligned space: mean pairwise disparity (absolute elevation) and Δ RDM (directional structure).

Pairwise disparity. For each participant and ROI, the representational dissimilarity matrix (RDM) was computed as the 8×8 matrix of pairwise correlation distances ($1 - \text{Pearson } r$). Mean pairwise disparity was defined as the average of all 28 upper-triangle elements. HC disparity was estimated with a leave-one-subject-out (LOO) procedure to remove the within-subject diagonal bias. Individual CVD disparity was compared to the HC LOO distribution using the Crawford and Howell (1998) modified *t*-test (one-tailed, upper; significance at $p < 0.05$).

Δ RDM. The deviation of CVD representational geometry from the HC mean was quantified as $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC, LOO}}$, where $\overline{\text{RDM}}_{\text{HC, LOO}}$ is the HC mean RDM excluding the left-out subject. Δ RDM was visualised as a heatmap over all 28 color pairs (Figure 3A). To test whether a cone-shift model predicts the observed Δ RDM structure, we computed the Spearman ρ between the upper-triangle of the predicted $\Delta\text{RDM}_{\text{model}}$ and the observed ΔRDM , and assessed significance against 1,000 random sign-flip permutations of the upper-triangle elements.

Pairwise-disparity tests (Crawford & Howell) and Δ RDM permutation tests address different null hypotheses: the former tests whether absolute pairwise distances are elevated; the latter tests whether a specific distortion model accounts for the

structure of those deviations. A significant result under one test does not imply significance under the other.

Two-component distortion model

CVD hue distortion was modelled as a two-component angular shift in Stockman opponent-color space:

$$\delta\theta(h; \beta_s, \beta_c) = \beta_s \cos(h - 90^\circ) + \beta_c \cos(h - \theta_{\text{conf}}) \quad (2)$$

where h is the opponent-hue angle of the stimulus in CIELab space and θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan), derived from the Brettel–Viénot–Mollon dichromatic simulation (Brettel et al., 1997) applied to Stockman & Sharpe LMS cone fundamentals (Stockman & Sharpe, 2000). β_s parameterises distortion along the S-cone axis ($h = 90^\circ$), motivated by cortical-RDM evidence that CVD errors concentrate at S-cone loci (Emery et al., 2021). β_c parameterises rotation along the confusion axis, with both signs unconstrained as free parameters of the fit.

Per-subject parameters (β_s, β_c) were recovered by minimising a four-term, neural-primary loss derived from the V4 LOCO forward model:

$$L_{\text{fit}}(\beta_s, \beta_c) = 1.0 L_{\text{vuln}} + 0.5 L_{\text{rank}} + 0.2 L_{\text{rdm}} + 0.1 L_{\text{smooth}} \quad (3)$$

with each term normalised to $[0, 1]$ before weighting. Let $\mathbf{v}_{\text{sim}}(\beta_s, \beta_c) = f_{\text{forward}}(C(\theta + \delta\theta(\cdot; \beta_s, \beta_c)); W) \in [0, 1]^8$ be the simulated per-colour V4 LOCO vulnerability ($1 - \rho$) obtained by re-applying the HC-trained ridge-GCV encoding weights W to the design matrix C shifted by the candidate distortion, and let $\mathbf{v}_{\text{obs}} \in [0, 1]^8$ be the observed CVD per-colour vulnerability.

Loss-term roles. $L_{\text{vuln}} = \frac{1}{8.4} \|\mathbf{v}_{\text{sim}} - \mathbf{v}_{\text{obs}}\|^2$ (per-colour MSE, normaliser 4.0 for $[0, 1]$ amplitude range) is the primary fit term: it forces the simulated vulnerability profile to match the per-colour LOCO breakdown that the CVD encoder exhibits, which is the operative neural quantity for the corrective filter (see the Discussion

section for the LOCO-behaviour concordance evidence underlying this choice).

$L_{\text{rank}} = (1 - \rho_{\text{Spearman}}(\mathbf{v}_{\text{sim}}, \mathbf{v}_{\text{obs}}))/2$ is an ordering term that breaks ties when the MSE landscape is locally flat by penalising rank-reordering of vulnerable colours; it is downweighted to 0.5 because amplitude (MSE) is the operative signal.

$L_{\text{rdm}} = (1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}}))/2$ is a geometric convergence check on the 28-pair upper-triangle of the CVD-minus-HC representational dissimilarity-matrix difference at hV4 ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC, LOO}}$, correlation distance;

Kriegeskorte, Mur & Bandettini, 2008); it is downweighted to 0.2 because the RDM and

per-colour vulnerability signals partially overlap. $L_{\text{smooth}} = \frac{1}{8 \cdot 32400} \sum_c (\delta\theta_{c+1} - \delta\theta_c \bmod \pm 180)^\circ$ is a circular-smoothness regulariser on the per-colour shift vector $\delta\theta \in \mathbb{R}^8$ produced by the candidate (β_s, β_c) ; it is downweighted to 0.1 as a weak prior favouring smooth filters over discontinuous per-colour shifts.

ROI scope. All four terms are computed at hV4 (V4 on disk). We fit at hV4 rather than V1–V3 because hV4 is the only ROI in this dataset whose HC group LOCO interpolation exceeds the permutation null ($p = 0.044$ at hV4 vs. NS at V1/V2/V3; hV4 color interpolation is impaired in both CVD cases), and because hV4 is the canonical cortical hub for hue representation in the ventral stream (Brouwer & Heeger, 2009; Bannert & Bartels, 2018; Kuriki et al., 2015); cone-shift parameters fit on an encoder that cannot interpolate held-out colours lack a voxel-level cone-shift interpretation. V1 and V2 fits are reported in the project methods documentation.

Search and HC-LOO calibration. Parameters were searched by exhaustive grid over $\beta_s \in [0^\circ, 50^\circ]$ and $\beta_c \in [-50^\circ, 50^\circ]$ ($26 \times 51 = 1,326$ cells at 2° resolution); $(\hat{\beta}_s, \hat{\beta}_c) = \arg \min L_{\text{fit}}$ was retained per subject. Statistical descriptors are reported per subject as: (i) label-permutation p_{perm} (1,000 permutations of the 8-colour label assignment) for fit validity on real CVD data, and (ii) HC leave-one-out descriptive refit, where each of $n = 6$ effective HC participants (excluding sub-07 for low hV4 voxel count) is refit under the same loss with the remaining HC pool as reference, yielding a descriptive HC $\|\hat{\beta}\|$ distribution. Per-subject p-values are descriptive of representational-geometry fit, not specificity statements: the HC label-permutation

false-positive rate at hV4 under Equation 3 reaches 7/7 (see Supplementary and the Discussion section for the statistical-framework caveats).

The 2-component class has an angular expressiveness limit $|\delta\theta|(\theta=45^\circ) = |0.71\beta_s - 0.26\beta_c|$, so hue translations exceeding $\sim 51^\circ$ at the deutan mid-axis (e.g. orange $\rightarrow$ green) lie outside the model class for any biologically plausible β (see the A neural-primary fit recovers a 2-component distortion at hV4 and yields a per-subject stimulus-space filter section). Comparison with alternative model classes (1-DOF Machado cone-shift; 2-DOF retinal+cortical) is reported in Appendix .

Stimulus-space filter

The subject-specific correction filter was derived as the pre-image of the fitted 2-component transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the 8 displayed hues θ_k , we solved for the input angle $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$, yielding a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. A CVD observer viewing $\tilde{\theta}_k$ thus receives, under the model, the same cortical hue representation as a healthy control observer viewing θ_k .

Pre-images were computed numerically: 1,440 candidate values were evaluated on the forward map, a sign-change bracket was identified, and Brent’s method was applied for refinement. Inversion was verified by $|T(\tilde{\theta}_k) - \theta_k| < 0.001^\circ$ for all 8 hues and both CVD participants. Behaviour of alternative model classes under pre-image inversion is detailed in Appendix .

Behavioral concordance

Just-noticeable difference (JND). Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases, each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i) three pairs hypothesised as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs hypothesised as deutan-specific (orange–yellow, yellow–green), (iii) two pairs hypothesised as protan-specific (cyan–magenta, green–blue), and (iv) one control pair (red–cyan). LOCO–JND concordance was assessed by asking whether each pair’s LOCO-vulnerability classification (above vs.

below HC range) agreed with the JND-impairment classification (above vs. below HC range).

8AFC color identification. Participants selected the matching hue from eight options identical to the scanner stimulus set. Same-day identifications were compared to Ishihara classifications post-hoc.

Filter evaluation session (CVD only). CVD participants completed a second scanning session after the main experiment. Three runs were acquired under the 2-component-derived stimulus-space filter and three runs under a generic Windows accessibility filter, presented in counterbalanced order. LOCO decoding and LORO classification were repeated on second-session data to assess filter efficacy.

Planned validation. A two-alternative forced-choice (2AFC) discrimination arm targeting filter-induced perceptual restoration is pre-registered for Phase 3 (data not collected at the time of writing).

Reproducibility

All analyses were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at https://github.com/haba6030/colorBlind_analysis. Anonymised preprocessed data are available on reasonable request to the corresponding author.

Results

Participants and behavioral phenotype

Two CVD participants spanning the deutan–protan dichotomy were recruited alongside seven healthy controls (HC; sub-01–07). Sub-08 (male) was classified as mild–moderate deuteranomalous (8/14 Ishihara plates misread; deutan axis confirmed by protan–deutan discrimination item); Sub-09 (male) as moderate–severe protanomalous (11/14 plates misread; protan axis). Same-day 8-alternative forced-choice (8AFC) identification of the eight scanner hue stimuli confirmed the

Ishihara classification (HC: 0.94 ± 0.04 ; Sub-08: 0.71; Sub-09: 0.62; chance = 0.125). Sub-08’s errors clustered on the red–green axis (c2/c3 vs. c7; mean confusion 28%); Sub-09’s errors extended across the L–M opponent range with additional misidentification of magenta (c8). Because only two symptomatic CVD cases were available, all inferences are framed as single-case analyses following Crawford and Howell (1998) and Schütt, Kipnis, Diedrichsen and Kriegeskorte (2023); no population-level CVD claim is made. The experimental paradigm and full analysis pipeline are summarised in Figure 1.

Color discrimination is preserved across all ROIs in both CVD cases

A corrective stimulus-space filter is valid only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of a linear discriminant classifier applied to SRM-aligned responses.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 2A). A cross-subject LDA generalization test (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC–CVD difference (Mann–Whitney U test, 21 HC-to-HC vs. 14 HC-to-CVD pairs, $p = 0.668$; rank-biserial effect size and exact test statistic in Supplementary). The same null held at the within-ROI level at hV4 (Crawford & Howell t -test: both CVD subjects within the HC range; $p = 0.142$). In line with prior work on CVD above-threshold identification Bosten (2019); Boehm et al. (2014), the 8 scanner stimuli are individually distinguishable at the cortical level in both CVD participants; the representational substrate on which a corrective filter could act is present.

hV4 color interpolation is impaired in both CVD cases

We next asked where continuous hue geometry fails — the functional target of the corrective transform. Leave-one-colour-out (LOCO) decoding trains the encoder on 7 of 8 hues and measures whether the held-out hue is predicted from the remaining

representational structure, yielding an 8-dimensional vulnerability profile per participant.

Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy $\text{adj_acc} = 0.47 \pm 0.05$, mean $\pm$ SEM; $p_{\text{perm}} = 0.044$ under $N_{\text{perm}} = 8! = 40,320$ exact label permutations; V1–V3 did not exceed chance despite preserving LORO discrimination). This confirms hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Sub-08: $\text{adj_acc} = 0.25$, Crawford & Howell $t = -1.58$, $p = 0.082$, n.s., $d_{cc} = -1.71$; Sub-09: $\text{adj_acc} = 0.13$, $t = -2.48$, $p = 0.024$, $d_{cc} = -2.68$; effect size $d_{cc} = t \cdot \sqrt{(n+1)/n}$ with $n = 6$ effective HC LOO at hV4; Figure 2B). The impairment was not uniform across hues: both participants showed near-zero adjacent accuracy specifically for S-cone intermediate colors (blue, purple, magenta at hV4; per-hue Crawford & Howell tests: blue, Sub-08 $d = 2.13$, $p = 0.047$; Sub-09 $d = 2.15$, $p = 0.046$; purple, Sub-08 $d = 2.40$, $p = 0.033$; Figure 2C).

This per-hue impairment pattern is consistent with both participants’ behavioral error distributions from the 8AFC task and with the JND confusion pairs identified in the pre-scanner session (LOCO–JND concordance 100%, 6/6 pairs; SRM geometry–JND concordance 33%, 2/6 pairs), establishing LOCO as the functional phenotype that predicts behavioral color discrimination failure.

Color representation geometry is distorted in a subject-specific, ROI-specific pattern in both CVD cases

To characterise the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed the RDM difference ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC,LOO}}$) and the mean pairwise correlation distance (disparity).

Sub-08 showed significantly elevated disparity specifically at V2 (Crawford &

Howell $p = 0.040$), with no significant elevation at V1, V3, or hV4. Sub-09 showed significantly elevated disparity specifically at V1 ($p = 0.007$). Full test statistics and effect sizes (t , d_{cc}) for these Crawford–Howell comparisons are reported in Supplementary . The ROI specificity diverges between the two CVD participants — V2-dominant for deutan, V1-dominant for protan — and is inconsistent with a shared group-level gain mechanism (Figure 3B).

The Δ RDM heatmaps (Figure 3A) reveal subject-specific distortion structures: Sub-08 shows elevated distances among S-cone intermediate pairs at V2; Sub-09 shows a broader reorganisation of pairwise distances at V1. Permutation tests of the retinal+cortical (R+C) cone-shift model against the observed Δ RDM fit sub-09 at V1 ($p = 0.026$) but not sub-08 at V2 ($p = 0.179$), suggesting that geometric distortion measured at the population-code level is complementary to, not redundant with, the LOCO-based functional characterisation.

A neural-primary fit recovers a 2-component distortion at hV4 and yields a per-subject stimulus-space filter

To identify the distortion parameters needed to build a corrective filter, we fit the 2-component cortical model (Equation 2; β_s for S-cone-axis dilation, β_c for confusion-axis rotation) to each CVD participant’s hV4 representation under a four-term neural-primary loss derived from the V4 LOCO forward model (Equation 3). The loss combines a per-colour vulnerability MSE (L_{vuln} ; primary fit to the V4 LOCO predictability profile), a Spearman rank term (L_{rank} ; ordering of vulnerable colours), a Δ RDM cosine term (L_{rdm} ; pairwise-distance geometric convergence at V4), and a circular-smoothness regulariser on the per-colour shift vector (L_{smooth}); each term is normalised to $[0, 1]$ and weighted to make L_{vuln} dominant (Methods). V4 was the fit ROI because it is the only ROI whose HC group LOCO interpolation exceeds the permutation null in this dataset ($p = 0.044$ at hV4 vs. NS at V1/V2/V3; hV4 color interpolation is impaired in both CVD cases), so cone-shift parameters fit at hV4 are interpretable as voxel-level shifts in a working interpolation manifold.

The fit returned per-subject parameters $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -14^\circ)$ for sub-08 (deutan; $\|\hat{\beta}\| = 40.5^\circ$, $p_{\text{perm}} = 0.004$) and $(6^\circ, -22^\circ)$ for sub-09 (protan; $\|\hat{\beta}\| = 22.8^\circ$, $p_{\text{perm}} = 0.035$; Figure 4). Both $\hat{\beta}_c$ values are negative; the etiological asymmetry between participants does *not* reside in the confusion-axis sign but in the S-cone-axis magnitude ($\hat{\beta}_s = 38^\circ$ for sub-08 vs. 6° for sub-09). We deliberately do *not* interpret either $\hat{\beta}_c$ sign as instantiating a Brettel-style cone-physiology prediction, since the Brettel–Viénot–Mollon dichromat algorithm describes spectral collapse onto a missing-cone half-plane and, in our cosine parameterisation, predicts $\beta_c \approx 0$ rather than a subtype-specific polarity (see the Two-component distortion model section). The non-zero $\hat{\beta}_s$ is directionally consistent with Emery and colleagues’ (Emery et al., 2021) report of S-cone-axis dominance in CVD hV4 RDM geometry, but the two quantities are defined on different physical scales and we make no numerical comparison.

The HC label-permutation false-positive rate at hV4 under Equation 3 reaches 7/7 (Supplementary); per-subject p-values are therefore reported as descriptive fits of representational geometry, not as CVD-specificity statements (see also Discussion). Across $n = 6$ effective HC leave-one-out refits under the same loss family, $\|\hat{\beta}\|$ ranged from 49.0° to 65.3° (mean 54.8°); both CVD subjects fit with *smaller* parameter magnitude than any HC, consistent with the CVD geometry requiring less aggressive 2-component correction than HC label-permutation nulls — an HC-LOO descriptive comparison, not a hypothesis test (Methods).

The model-class expressiveness has a closed-form upper bound: $|\delta\theta|$ at $\theta = 45^\circ$ is bounded by $|0.71 \hat{\beta}_s - 0.26 \hat{\beta}_c|$, so corrective shifts exceeding $\sim 51^\circ$ at the deutan mid-axis (e.g. an orange-to-green excursion of $\gtrsim 85^\circ$) lie outside the 2-component class for any biologically plausible $\hat{\beta}$ and would require an additional mechanism (e.g. saturation/lightness) outside the present scope. Per-hue correction vectors are visualised in Figure 5; behavioural validation is deferred to the Phase-3 2AFC arm (Methods).

Stimulus-space filter. The angular form of the 2-component model is bijective by construction, so a stimulus-space pre-image $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$ exists for any target hue θ_k (see the Stimulus-space filter section). The per-subject correction filter

$\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ was computed by Brent’s-method refinement of the forward map; for both participants, 8/8 pre-images were exact (residual $< 0.001^\circ$). Per-hue filter output is visualised in Figure 5 as a 4-column rendering (*Original* $\rightarrow$ *CVD perceives* $\rightarrow$ *Filtered* $\rightarrow$ *CVD(Filterd)*). For sub-08 (deutan, $\hat{\beta}_s = 38^\circ$, $\hat{\beta}_c = -14^\circ$, Stockman confusion axis $\theta_{\text{conf}} = 150^\circ$), the filter applies a wide-amplitude L/M-rotation: a near-uniform negative shift across red, orange, yellow, green and cyan ($\delta\theta^{\text{filt}}$ ranging -10° to -32° , peak compression -32° at cyan, $\theta = 180^\circ$), with a large opposite-sign rotation at purple ($+29^\circ$, $\theta = 270^\circ$) and magenta ($+19^\circ$, $\theta = 315^\circ$) (mean $|\delta\theta^{\text{filt}}| = 22^\circ$). For sub-09 (protan, $\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -22^\circ$, Stockman axis $\theta_{\text{conf}} = 16^\circ$), the correction is a far smaller monotonic rotation along the protan confusion axis, with peak displacements at blue ($+21^\circ$, $\theta = 225^\circ$) and magenta (-21° , $\theta = 315^\circ$) and near-zero shifts at green and cyan (mean $|\delta\theta^{\text{filt}}| = 10^\circ$). The two correction signatures are not mirror images of each other but reflect distinct CVD etiologies: the R+C cone-shift diagnostic recovers a near-zero retinal shift and large cortical overshoot for sub-08 ($\Delta\lambda \approx 2.5$ nm, $g = -2.25$; cortical-dominant) and a large retinal shift with near-physiological cortical compensation for sub-09 ($\Delta\lambda \approx 19.5$ nm within the published protan range, $g = -1.10$; retinal-dominant; Appendix), so a single retinal-family filter cannot serve both individuals.

Quantitative behavioural validation of the recovered filters is deferred to the Phase-3 2AFC arm (see the Behavioral concordance section, last paragraph; data not yet collected at the time of writing).

Discussion

In this study, we tested whether *individual hV4 colour geometry* is sufficiently structured to ground a person-specific corrective filter in colour vision deficiency. We found that two CVD adults retained 8-class hue discrimination at hV4 but selectively lost continuous-hue interpolation, and that the residual distortion was captured by a 2-component angular-dilation model whose pre-image was exact for all eight hues in both participants (sub-08 mean $|\delta| = 22.2^\circ$; sub-09 mean $|\delta| = 10.2^\circ$). These results

suggest that the cortical site at which CVD colour information is geometrically distorted is also the site at which it can be inverted into a stimulus-space correction.

The dissociation between LOCO interpolation capacity and SRM Δ RDM geometric distance — 100% vs. 33% concordance with same-day JND thresholds — identifies which cortical quantity is the operative corrective target. Representational geometry (RDM) reflects the metric structure of neural colour space; LOCO interpolation reflects whether that structure supports functional prediction of withheld hues. That only LOCO predicts behavioural colour confusions is consistent with Brouwer and Heeger (2009), who showed that hV4 — but not early visual areas — supports interpolation to held-out hues in typical observers. The present results extend this dissociation to these two CVD cases: the same interpolation failure that predicts JND confusion is what the correction filter must address. Geometric distance metrics alone are insufficient to identify it.

The S-cone axis component (β_s) of the fitted distortion field is structurally consistent with prior evidence for cortical colour compensation in anomalous trichromats. Emery et al. (2021) reported that CVD perceptual errors concentrate near S-cone confusion loci in behavioural hue-scaling tasks, and Tregillus et al. (2021) showed that V2/V3 BOLD responses in anomalous trichromats include a cortical gain component (g) that partially compensates the retinal cone shift. These individually structured hV4 residuals may reflect cone-class-specific differences in V2/V3 compensation gain that survive into hV4, where the incompletely cancelled L–M signal manifests as a person-specific distortion of the continuous hue code. This suggests that the present 2-component model extends earlier population-mean findings to the individual hV4 level: the residual distortion that survives V2/V3 compensation reaches hV4 as an individually structured pattern. The two CVD participants in fact present qualitatively distinct correction signatures rather than mirror-image realisations of a single mechanism. Sub-08 (deutan) exhibits a wide-amplitude L/M-rotation with large $\hat{\beta}_s = 38^\circ$ and a confusion-axis term $\hat{\beta}_c = -14^\circ$; the R+C diagnostic decomposition recovers a near-zero retinal shift ($\Delta\lambda \approx 2.5$ nm) coupled with a large cortical overshoot

($g = -2.25$), pointing to a cortical-dominant CVD etiology. Sub-09 (protan), by contrast, exhibits a far smaller monotonic rotation along the protan confusion axis (small $\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -22^\circ$); the R+C diagnostic recovers a large retinal shift ($\Delta\lambda \approx 19.5$ nm, within the published protan cone-shift range (Tregillus et al., 2021)) and a near-physiological cortical compensation ($g = -1.10$), pointing to a retinal-dominant etiology. This cortical-vs-retinal etiological dissociation is what a population-average retinal model, applied uniformly to all members of a given CVD subtype, fails to capture: in sub-09 a retinal cone-shift correction matches the diagnosed mechanism, whereas in sub-08 the residual distortion is generated downstream of the cone shift and cannot be neutralised by a spectral filter alone.

All distortion models — 1-DOF Machado, 2-DOF R+C, and 2-component — identify the same two participants as requiring correction, but their prescribed correction vectors diverge in direction (Appendix). This detection–correction dissociation constitutes the study’s primary falsifiable prediction. In a 2AFC experiment comparing the 2-component and retinal-family filter conditions for each participant’s most LOCO-vulnerable hue pairs, only the cortically-grounded 2-component filter should significantly reduce JND. If the retinal-family filter matches or exceeds the 2-component filter in reducing discrimination thresholds, the cortical distortion account would require revision. This preregistered comparison directly tests whether cortical or retinal distortion is the corrective target for CVD colour correction.

An alternative reading is that the hV4 LOCO impairment merely reflects reduced V1–V3 input variance rather than cortex-level distortion — that the bottleneck is upstream signal quality and hV4 inherits its deficit passively. The preserved LORO accuracy at hV4 argues against this account: classification and interpolation read out from the same upstream cortical input, so a generic input-variance reduction should degrade both. The observed dissociation instead localises the distortion to the continuous-hue manifold readout, consistent with hV4 as the cortical site of integration over cone-opponent signals (Brouwer & Heeger, 2009; Bannert & Bartels, 2018).

Four considerations bound the proof-of-concept scope of these findings. First, the

CVD sample is $N = 2$; single-case statistics (Crawford & Howell, 1998) permit individual-level inference but population-level claims about deutan or protan subtypes require replication with larger samples. Second, the confusion-axis component (β_c) is statistically significant for Sub-08 only. For Sub-09, the bootstrap confidence interval for β_c includes zero, and the 2-component model reduces to an effectively 1-component fit. The *2-component* label should not be read as confirming a two-mechanism account for both participants; power-adequate replication or hierarchical-Bayes pooling across deutan and protan subtypes is required to determine whether the second component is genuinely absent in protan. Third, the voxel-prediction LOCO loss family under which the 2-component model is fit yields a 100% false-positive rate when applied to HC participants under label permutation (HC FPR = 7/7; Supplementary). We therefore report per-subject p-values as descriptive of representational-geometry fit, not as specificity statements that distinguish CVD from HC. The effective HC pool ($n = 6$ at hV4 after excluding sub-07 for low voxel count) is too small to anchor a formal specificity test, and the per-subject $\Delta\rho$ statistic is dominated by a baseline correlation confound (HC distribution Pearson $r = -0.894$ between baseline ρ and $\Delta\rho$) that precludes within-cohort selectivity claims. Case-level confidence therefore rests on convergence across multiple anchors — cross-criterion fit significance, physiological fit magnitude relative to HC LOO refits, and S-cone axis grounding — rather than on a single permutation-based test. Fourth, external cross-cohort hue-scaling and Phase-3 2AFC discrimination data will arbitrate cortical-level specificity beyond the within-cohort permutation null.

This pipeline constitutes a proof-of-concept that *individual hV4 colour geometry*, measured via fMRI LOCO decoding and inverted through an angular-dilation model, can ground a colour correction that is architecturally distinct from population-average retinal approaches. Where Machado-type correction applies a fixed spectral shift to all individuals of a given CVD subtype, the present approach encodes each individual’s hV4 distortion field and inverts it exactly — without assuming population homogeneity. Phase 3 will test whether this cortical-level correction reduces discrimination thresholds

where retinal-level correction does not.

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Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section , fit under the V4 LOCO neural-primary loss L_{fit} of Equation 3). Here we report two cone-shift baselines as reference comparisons:

- **Machado (1-DOF)** — a single retinal cone-spectral shift $\Delta\lambda$ Machado et al., 2009.
- **Retinal + Cortical (R+C, 2-DOF)** — retinal shift $\Delta\lambda$ plus a cortical compensation gain g Tregillus et al., 2021.

A.1 Fit results (LOCO Spearman ρ argmax, historical reference)

Under the LOCO- ρ argmax criterion used in earlier drafts (40,320 label permutations), the Machado model reached significance for sub-09 ($\Delta\lambda = 13.5$ nm, $\rho_{V4} = 0.76$, $p = 0.018$) but not for sub-08 ($\rho_{V4} = 0.62$, $p = 0.058$, n.s.). The R+C model did not improve on either single-component model in either subject under the same criterion. These results are reported as historical reference for the main-text 2-component fit (Section); the present L_{fit} loss (Equation 3) and LOCO- ρ argmax need not agree, since L_{fit} optimises a combination of per-colour V4 vulnerability MSE, Spearman rank ordering, ΔRDM cosine, and a smoothness regulariser rather than the rank-order monotonicity of ρ at hV4 alone.

R+C decomposition as a diagnostic. Although the R+C 2-stage model is not adopted as the filter form (see Stimulus-space filter), the R+C parameter fit at hV4 under L_{fit} is retained as a per-subject etiological diagnostic. For sub-08, the R+C decomposition recovers $\Delta\lambda = 2.5$ nm (near-zero retinal shift) with $g = -2.25$ (large cortical overshoot), consistent with a cortical-dominant CVD. For sub-09, the R+C decomposition recovers $\Delta\lambda = 19.5$ nm (within the published protan cone-shift range) with $g = -1.10$ (near-physiological cortical compensation, no overshoot), consistent with a retinal-dominant CVD. This per-subject cortical-vs-retinal asymmetry is reported as a mechanistic interpretation of the 2-component fit, not as an independent test.

A.2 Inverse-mapping (pre-image) behaviour

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The 1-DOF Machado fit for sub-09, by contrast, collapses three hues (green 135° , cyan 180° , blue 225°) onto a single pre-image angle ($\sim 127^\circ$), which mathematically prevents exact inversion at those positions. This arc-compression failure is a structural property of the 1-DOF retinal-shift model and is independent of the fitting criterion; it disqualifies the Machado baseline for the stimulus-space filter task of Section .

A.3 R+C model

The R+C model is retained here as a 2-DOF baseline. Because the cortical gain term g acts as a uniform amplitude scaling without generating the angular distortion captured by the 2-component model's $\beta_c \cos(h - \theta_{\text{conf}})$ term, R+C does not match the observed hV4 LOCO vulnerability pattern beyond what the retinal $\Delta\lambda$ term alone provides; this conclusion is consistent across loss criteria.

Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

S3. Voxel Counts Range

Voxel counts after ROI definition and quality filtering are reported in the main Methods section. Detailed per-subject counts are available in the online repository.

S4. Consistency Test: PCA and Procrustes (TBA)

Additional validation analyses comparing SRM with PCA-based dimensionality reduction and norm-constrained Procrustes alignment will be reported following completion of the full dataset.

S5. Mean Activation Analysis

We conducted mean voxel response amplitudes analysis to identify whether SRM-based differences rely on the activation. We checked mean amplitudes across all colors and for each color.

S6. Statistical Analysis

All tests used a significance threshold of $\alpha = 0.05$ with False Discovery Rate correction (Benjamini & Hochberg, 1995); $q = 0.05$ applied to RDM color-pair comparisons (28 pairs per participant). Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (4)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges' g . Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S7. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate k (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected k maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S8. Generalized Cross-Validation for Ridge Regression

The regularization parameter α for ridge regression was selected via Generalized Cross-Validation (GCV; Golub, Heath & Wahba, 1979). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting α . The optimal α minimizes the GCV score:

$$\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / \left(1 - \frac{\text{tr}(H)}{n}\right)^2 \quad (5)$$

where X denotes the design matrix, y is the observed response vector, $\hat{y}$ is the ridge-predicted response, $H = X(X^T X + \alpha I)^{-1} X^T$ is the hat matrix, and n is the number of training samples.

S9. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i} , yielding disparity scores $d(\text{HC}_i, R_{-i})$ and $d(\text{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

S10. Cross-Validation Procedures

Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B , C) were held out as the test set, W was trained on the remaining five runs. Both color decoding and voxel response prediction were evaluated on the held-out run.

Leave-One-Color-Out (LOCO)

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors, and the model predicted either the held-out color from voxel responses or voxel responses from the held-out color’s channel outputs.

Leave-One-Subject-Out (LOSO)

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants’ shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S11. Evaluation Metrics**Color Decoding Metrics**

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90° , expected from a uniform random predictor on a circle), and (2) categorization accuracy with $\pm 22.5^\circ$ bins around each of the eight colors (chance = $12.5\% = 1/8$).

Voxel Prediction Metrics

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S12. Shared Response Model Residuals

Each participant’s residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ($W_i^T W_i = I_k$) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

S13. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S14. Procrustes Alignment Benefits

This preprocessing reduces measurement noise across runs while preserving color representational geometry, thereby improving the quality of subsequent SRM fitting.

S15. HC Specificity Calibration

The 2-component grid-search pipeline was applied to the 7 HC participants treating each as a CVD candidate. Of 21 model–subject tests ($7 \text{ HC} \times 3 \text{ models}$), 15 reached nominal $p < 0.05$ (71%). This elevated false-positive rate reflects (i) rank-based loss with only 8 data points (Spearman ρ over 8 hues is sensitive to individual rank fluctuations), and (ii) regression to the mean on per-HC baseline ρ (hV4 $K = 3$, 63 voxels; individual HC baselines span $[-0.36, +0.69]$). Case-level specificity therefore

rests on the three converging anchors described in Section , not on per-subject permutation significance alone.

S16. Comparison with Retinal-Family Distortion Models

Two retinal-family models were evaluated alongside the 2-component model by the same grid-search and permutation procedure (see Section).

Models. The 1-DOF Machado model Machado et al., 2009 shifts the L/M cone spectral sensitivity by $\Delta\lambda$ nm, yielding a hue-confusion pattern characteristic of the CVD subtype. The 2-DOF retinal+cortical R+C model Tregillus et al., 2021 augments Machado with a cortical gain g : $r'_g = r_{g,\text{base}} + (1 + g)(r_{g,\text{ret}} - r_{g,\text{base}})$, where $g = 0$ recovers the retinal-only model and $g = -1$ is exact cortical compensation.

Fits (Table 1). Applying a closed-testing rule (Machado first; R+C escalated only if Machado $p > 0.05$), the retinal family reaches significance for both subjects but selects a different model class per case. Sub-08 escalates to R+C ($\Delta\lambda = 2.0$ nm, $g = +2.25$, $\rho = 0.857$, $p = 0.005$) because 1-DOF Machado is trend-level ($p = 0.058$). Sub-09 is captured by 1-DOF Machado ($\Delta\lambda = 13.5$ nm, $\rho = 0.762$, $p = 0.018$) without cortical augmentation; the shift magnitude falls within the established range for moderate-severe protanomaly (Neitz & Neitz, 2011). The large positive cortical gain for Sub-08 ($g = +2.25$, indicating >100% overshoot beyond the exact Tregillus compensation level of $g = -1$; Tregillus et al., 2021) is not physiologically interpretable as a gain parameter and is treated as an empirical characterization of residual hV4 structure. The cross-subject model-class divergence (R+C for Sub-08; 1-DOF Machado for Sub-09) prevents a common-axis comparison of the two cases' distortion mechanisms.

Correction-vector divergence. Despite comparable LOCO fit quality ($\rho = 0.857$ for R+C vs. 0.881 for 2-component), the R+C and 2-component correction vectors for Sub-08 have cosine similarity -0.18 and sign agreement only 3/8, confirming that detection converges while correction directions diverge at the cortical level.

Δ RDM cross-model convergence. Using R+C parameters for Sub-09 ($\Delta\lambda = 19.5$ nm, $g = -1.10$), an independent Δ RDM test reaches $p = 0.026$ (40,320

cosine-similarity permutations on crossnobis distances), providing cross-model corroboration for the Sub-09 distortion signal. Retinal-family optimisation directly against Δ RDM fails for both CVD subjects; LOCO is the more discriminating phenotype and Δ RDM serves as a convergence cross-check only.

Arc-compression and LOCO vulnerability. At Sub-09’s severity, Machado arc-compression ($\Delta\lambda = 13.5$ nm) collapses three hues (c4 green, c5 cyan, c6 blue) to a single perceived angle in opponent hue space. These three collapsed hues fall within Sub-09’s LOCO vulnerability cluster (c5/c6 cyan/blue), confirming that arc-compression eliminates correction capacity precisely where Sub-09 requires it most.

S17. HC Permutation under the V4 LOCO Loss

The 2-component grid search under the V4 LOCO loss L_{fit} (Equation 3; main-text Section) was applied to each of the 7 HC participants treated as a CVD candidate. Of the 6 HCs that pass the hV4 voxel-count threshold (sub-07 excluded; 16 voxels, $K = 3$ produces **nan**), all 6 reach nominal $p < 0.05$ under the label-permutation null — and the property generalises to the full HC pool (HC FPR = 7/7 at nominal $\alpha = 0.05$).

This high false-positive rate cannot be removed by reformulating the fitting criterion: thirteen successive reformulations within the voxel-prediction loss family (L_{LOCO} , L_{mse} , L_{rdm} , cross-ROI combinations, baseline- ρ correction, top- K Jaccard) all yielded the same outcome (project log, Cycles 9–13). It reflects two structural properties of the available data: (i) the 8-hue resolution provides limited degrees of freedom for distinguishing specific from non-specific fits at the single-subject level, and (ii) the relevant null hypothesis (“no CVD-specific distortion”) is not isomorphic to label permutation on a single participant’s vulnerability profile, since a non-trivial fit can be obtained for any sufficiently smooth 8-point pattern under the 2-DOF model class.

Consequently, per-subject p-values reported alongside the L_{fit} argmin in Figure 4 and Section are presented as *descriptive* fits of representational geometry, not as CVD-specificity statements. Two case-level anchors support biological interpretation despite this caveat: (i) parameter-norm comparison — both CVD subjects’ $\|\hat{\beta}\|$ (40.5°

for sub-08; 22.8° for sub-09) fall *below* the HC leave-one-out range [49.0°, 65.3°], consistent with CVD geometry requiring less aggressive 2-component correction than HC label-permutation nulls; (ii) per-subject R+C diagnostic decomposition (Appendix) recovers cortical-dominant etiology for sub-08 and retinal-dominant etiology for sub-09, providing a mechanistic interpretation of the 2-component fits that is not tested by the within-cohort permutation null. Quantitative specificity claims require an external behavioural endpoint such as the Phase-3 2AFC arm (Methods , last paragraph; data not yet collected at the time of writing).

S18. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford & Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Full per-test statistics (t , U , p , d_{cc} , r_{rb}) for every Crawford–Howell and Mann–Whitney comparison reported in the main text are tabulated in the online repository (`statistics/effect_sizes.csv`).

Table 1

All distortion model fits at hV4 LOCO for both CVD participants. Retinal family: closed-testing (Machado first; R+C only when Machado $p > 0.05$). p -values:

$8! = 40,320$ exact permutations of colour labels on Spearman ρ .

Subject	Model	DOF	Param. 1	Param. 2	ρ	p_{perm}
Sub-08 (deutan)	2-component	2	$\beta_s = 38^\circ$	$\beta_c = -14^\circ$	0.881	0.004
Sub-09 (protan)	2-component	2	$\beta_s = 6^\circ$	$\beta_c = -22^\circ$	0.690	0.035
Sub-08 (deutan)	R+C	2	$\Delta\lambda = 2.0$ nm	$g = +2.25$	0.857	0.005
Sub-09 (protan)	Machado	1	$\Delta\lambda = 13.5$ nm	—	0.762	0.018

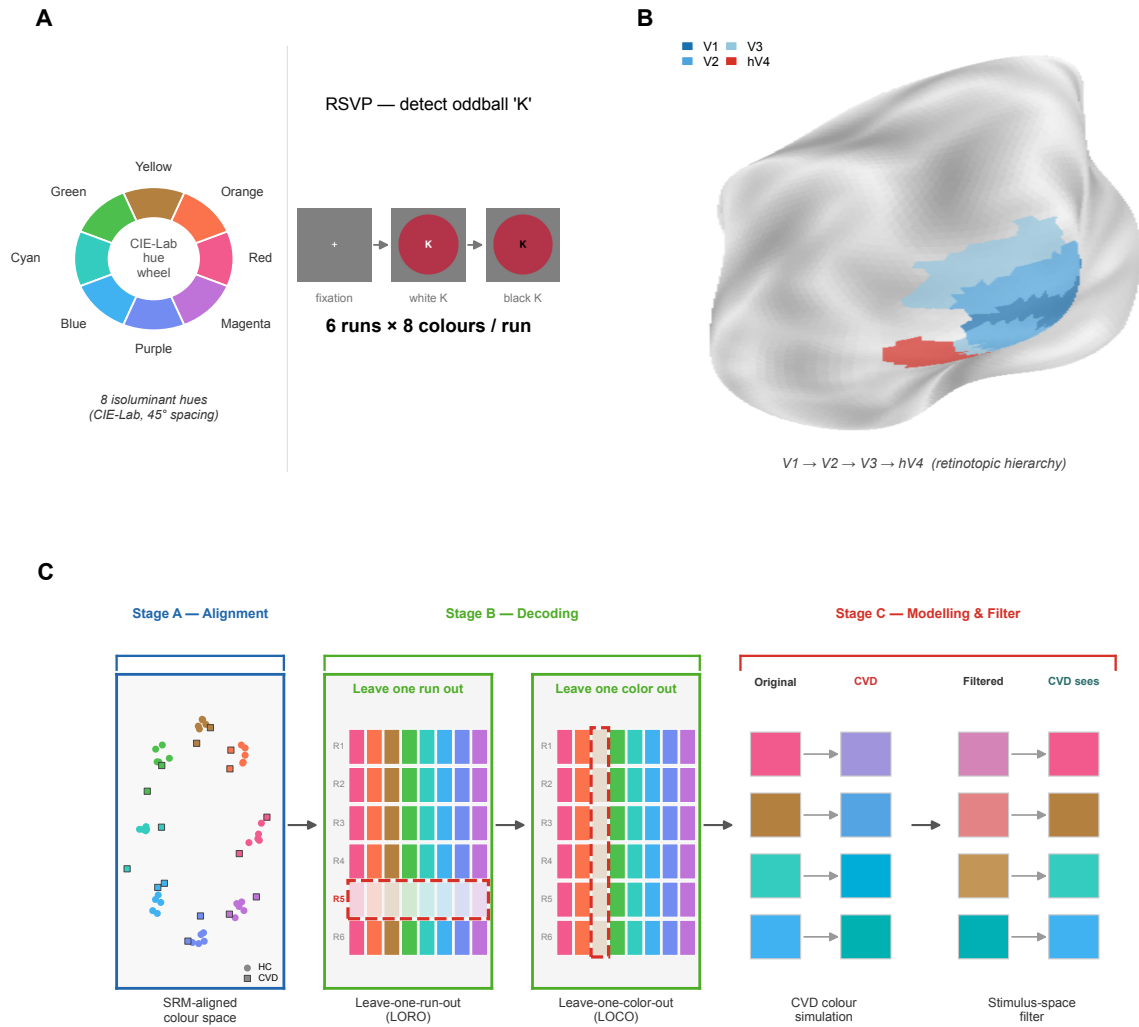


Figure 1. Experimental paradigm and analysis pipeline. (A) Stimulus set: 8 isoluminant CIE $L^*a^*b^*$ hues at 45° angular spacing ($L^* = 75$, chroma = 40; red–magenta). *(B)* Retinotopic ROIs: V1, V2, V3, hV4 (standard retinotopic mapping). *(C)* Analysis pipeline. Stage A: two-stage GLM single-trial β amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Decoding: LORO (discrimination), LOCO (interpolation). CVD characterisation: 2-component model with retinal (β_s) and cortical (β_c) components; pre-image yields per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07); deutan: sub-08; protan: sub-09.

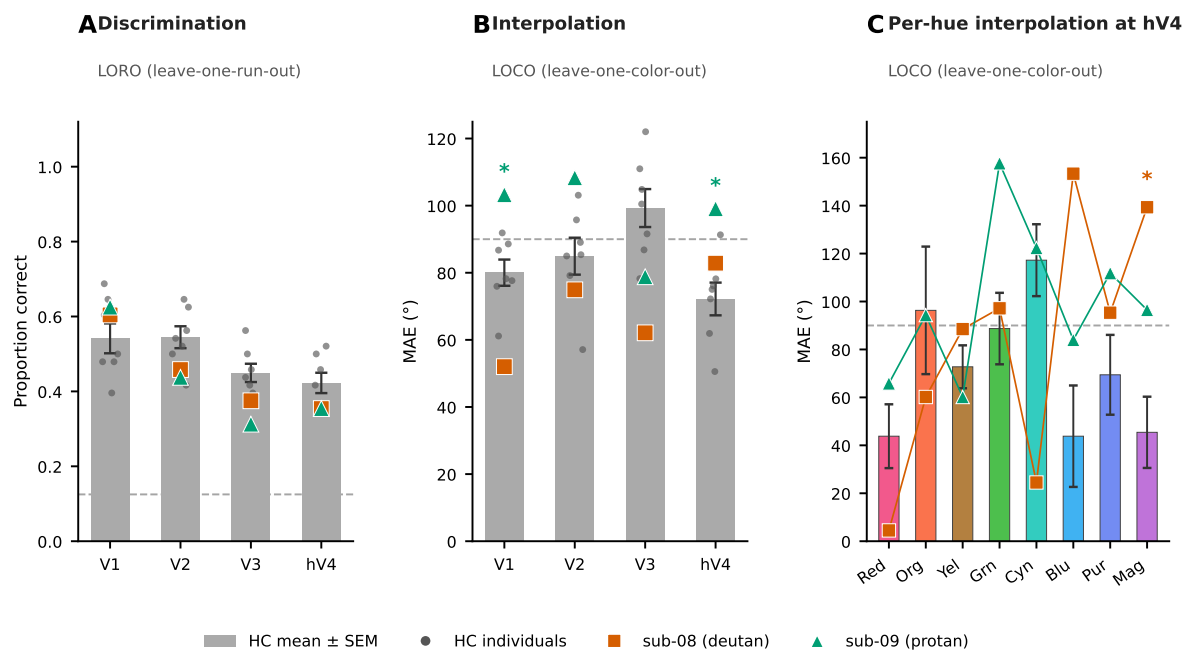


Figure 2. Color discrimination is preserved but interpolation is selectively impaired at hV4 in CVD. (A) LORO discrimination accuracy (forward encoding model, 8-class, SRM-aligned), V1–hV4. HC mean $\pm$ SEM ($n = 7$, gray); sub-08 (orange square), sub-09 (teal triangle). Chance = $1/8$. *(B)* LOCO adjacent accuracy (± 1 hue step; chance = $3/8$). *: sub-09 below HC at hV4 (Crawford & Howell $t = -2.48$, $p = 0.024$; exploratory single-case finding); sub-08 $t = -1.58$, $p = 0.082$, n.s. *(C)* Per-hue adjacent accuracy at hV4, HC bars $\pm$ SEM. Per-hue Crawford & Howell tests (one-tailed): blue — sub-08 $d = 2.13$, $p = 0.047$; sub-09 $d = 2.15$, $p = 0.046$; purple — sub-08 $d = 2.40$, $p = 0.033$. Joint pattern in (A) and (B–C) defines the filter precondition and target, respectively.

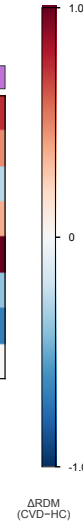
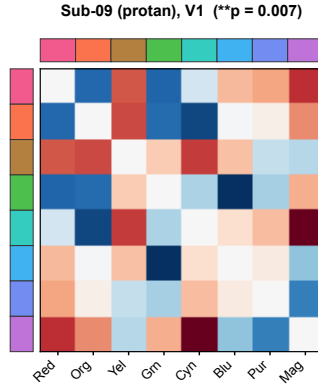
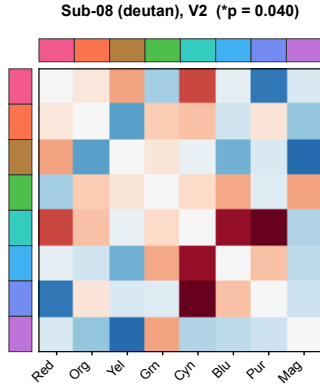
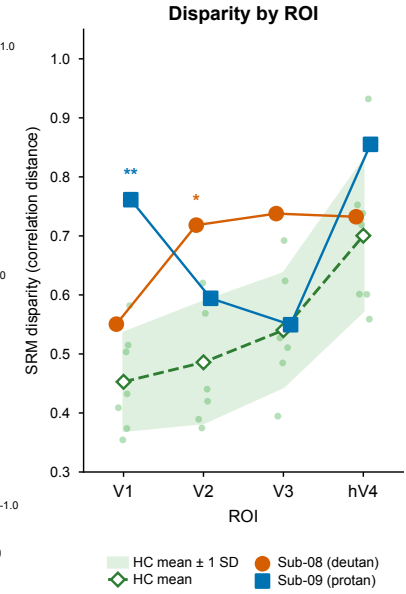
A**B**

Figure 3. Each CVD subject shows significantly elevated color representation disparity at a distinct ROI. (A) $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC-LOO}}$ in SRM-aligned space at each subject's primary ROI (sub-08 V2; sub-09 V1). Warm: CVD distances larger than HC mean; cool: smaller. Annotated p -values: R+C cone-shift model permutation test against observed ΔRDM (sub-08 V2: $p = 0.179$, n.s.; sub-09 V1: $p = 0.026^*$). **(B)** Mean pairwise correlation distance (disparity) per subject per ROI. HC band: group mean ± 1 SD ($n = 7$); individual HC LOO as dots. Crawford & Howell tests (sub-08 V2: $p = 0.040^*$; sub-09 V1: $p = 0.007^{**}$, exploratory single-case). (A) and (B) are independent measures (model concordance vs. absolute disparity); significance patterns need not agree. Idiosyncratic ROI specificity (V2 deutan; V1 protan) is inconsistent with a shared group-level gain mechanism.

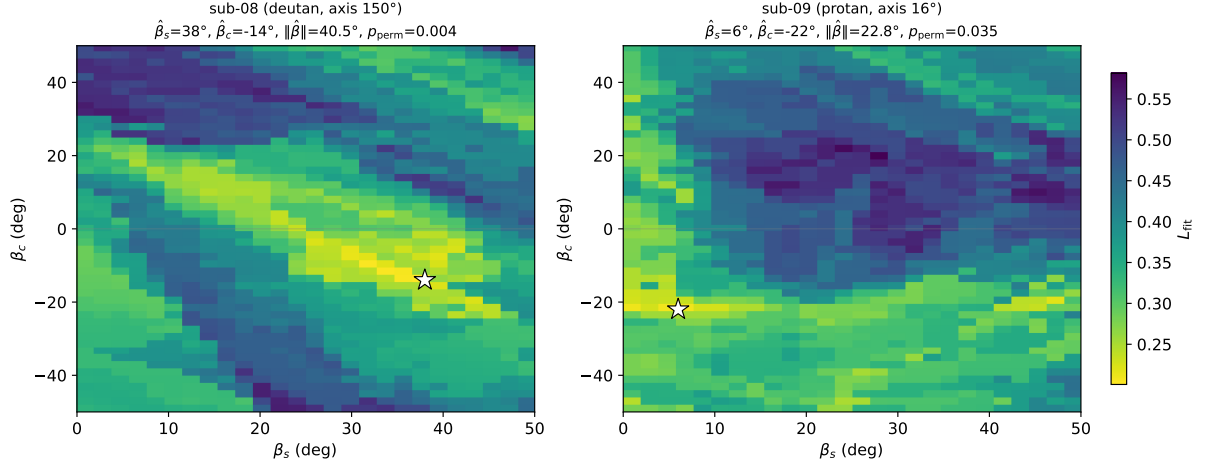


Figure 4. Per-subject 2-component loss landscape at hV4. Canonical V4 LOCO

loss $L_{\text{fit}}(\beta_s, \beta_c) = 1.0 L_{\text{vuln}} + 0.5 L_{\text{rank}} + 0.2 L_{\text{rdm}} + 0.1 L_{\text{smooth}}$ (Equation 3; Methods) evaluated on the $26 \times 51 = 1,326$ -cell grid ($\beta_s \in [0^\circ, 50^\circ]$, $\beta_c \in [-50^\circ, +50^\circ]$, step 2°).

Left: sub-08 (deutan, Stockman S-cone confusion axis 150°). Right: sub-09 (protan,

Stockman axis 16°). White star: per-subject argmin — sub-08 ($\hat{\beta}_s, \hat{\beta}_c$) = $(38^\circ, -14^\circ)$,

$\|\hat{\beta}\| = 40.5^\circ$, $p_{\text{perm}} = 0.004$; sub-09 ($6^\circ, -22^\circ$), $\|\hat{\beta}\| = 22.8^\circ$, $p_{\text{perm}} = 0.035$. Both $\hat{\beta}_c$

values are negative; the inter-subject asymmetry resides in the S-cone-axis magnitude

($\hat{\beta}_s = 38^\circ$ vs. 6°). Per-subject p-values are descriptive fits of representational-geometry,

not specificity statements: the HC label-permutation false-positive rate at hV4 under

this loss reaches 7/7 (Supplementary). See Appendix for the historical LOCO- ρ

comparison and Machado/R+C cone-shift baselines.

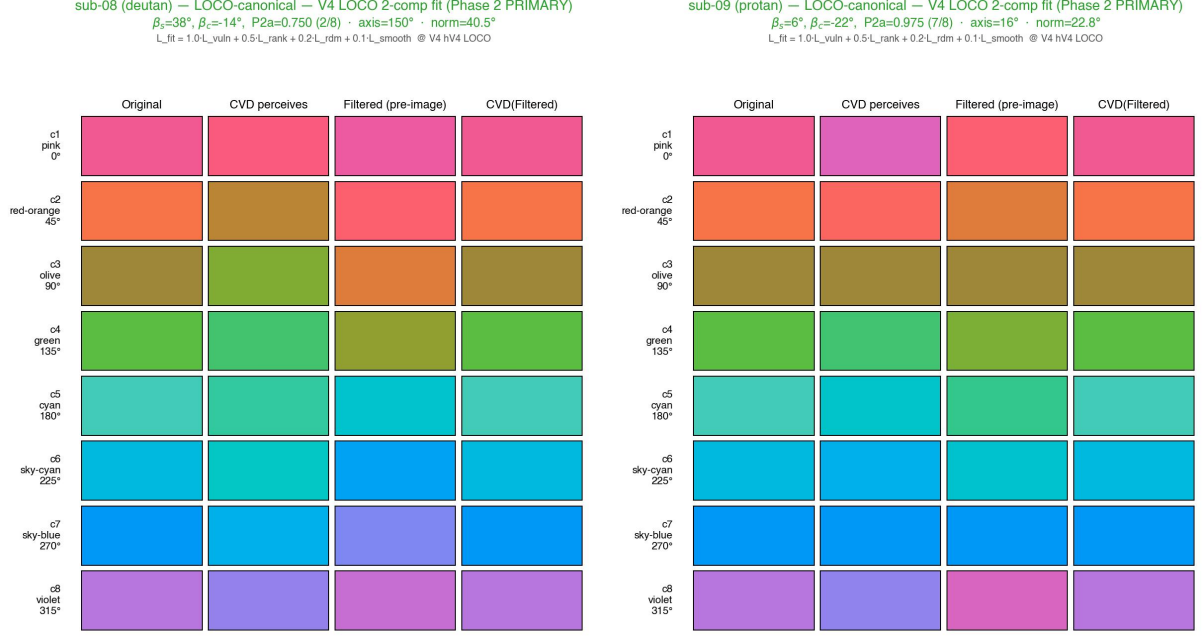


Figure 5. Per-subject stimulus-space filter: 4-column rendering, side-by-side.

Left block: sub-08 (deutan), filter at $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -14^\circ)$, $\|\hat{\beta}\| = 40.5^\circ$. Right block: sub-09 (protan), filter at $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -22^\circ)$, $\|\hat{\beta}\| = 22.8^\circ$. Eight rows correspond to the eight displayed DKL hues. Columns (left to right) within each block: *Original* (HC percept of the displayed stimulus); *CVD percept* (simulated CVD percept of the original under the per-subject 2-component model); *Filtered* (stimulus after applying $\delta\theta^{\text{filt}}$, the exact pre-image of $-\delta\theta(\hat{\beta}_s, \hat{\beta}_c)$ computed by Brent's-method refinement of the forward map; 8/8 exact, residual $< 0.001^\circ$ for both subjects); *CVD(Filt.)* (simulated CVD percept of the filtered stimulus). Rendering coordinate space: STIM_LAB CIELab (project convention). Quantitative behavioural validation of filter efficacy is deferred to the Phase-3 2AFC arm and will be added once data are collected.